2025_Hexaware_Actual_Test_13

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. A can finish a work in 8 hours, B can finish the same work in 12 hours, and C can do it in 24 hours. All the three starts work at 3 am, 4 am, and 5 am respectively. At what time work will be completed?

- 6:30 am
- 7: 40 am
- 8: 20 am
- 5: 50am

Q2. P, Q, and R get Rs.1105 for doing a job together. While working alone, P, Q, and R need 7 days, 8 days, and 11 days respectively to complete the job. Then what would be the share of each?

- P = Rs. 440; Q = Rs. 385; R = Rs. 280
- P = Rs. 500; Q = Rs. 400; R = Rs. 205
- P = Rs. 445; Q = Rs. 375; R = Rs. 285
- P = Rs. 446; Q = Rs. 381; R = Rs. 278

Q3. An athlete runs a 200 m race in 24 seconds. His speed (kmph) is

- 24
- 20
- 30
- 28

Q4. A steamer running downstream covers a distance of 30 km in 2 hours. While coming back the steamer takes 6 hours to cover the same distance. If the speed of the current is half of that of the steamer, then find the speed of the steamer in kmph

- 2 kmph
- 5 kmph
- 10 kmph
- 15 kmph

Q5. A shopkeeper gives 3 consecutive discounts of 10%, 15% and 15% after which he sells his goods at a percentage profit of 30.05% on the C.P. Find the value of the percentage profit that the shopkeeper would have earned if he had given discounts of 10% and 15% only.

- 53%
- 62.5%
- 72.5%
- 68.6%

Q6. A discount series of 10%, 20%, and 40% is equivalent to a single discount of ?

- 56.8%
- 55%
- 62%
- 57.5%

Q7. Deepa in her lab finds a 90kg bag of mixture which contains 90% sugar in it. This mixture is made not only of sugar but also has got maida in it. She now adds another 30kg of sugar to the bag of mixture. Now, how much percentage of sugar will be present in the new mixture?

- 45 $\frac{1}{2}$
- 60 $\frac{1}{5}$
- 43 $\frac{1}{2}$
- 92 $\frac{1}{2}$

Q8. The price of Dal is increased from 30 to 37.5 per kg. Find the percentage increase in price.

25

60

45

None

Q9. A sum of money placed at compound interest doubles itself in 3 years. In how many years will it amount to 8 times itself?

9 years

8 years

27 years

7 years

Q10. Rs. 10000 is being compounded at 20% p.a. If the rate of interest is charged half-yearly. What will be the amount in Rs. after 2 years?

14000

14500

14641

15000

Q11. Alphonsa, on his death bed, keeps half his property for his wife and divides the rest equally among his three sons: Ben, Carl, and Dave. Some years later, Ben dies leaving half his property to his widow and half to his brothers Carl and Dave together, sharing equally. When Carl makes his will, he keeps half his property for his widow, and the rest he bequeaths to his younger brother Dave. When Dave dies some years later, he keeps half his property for his widow and the remaining for his mother. The mother now has Rs 1,575,000. What was the worth of the total property?

Rs.30 lakh

Rs.8 lakh

Rs.18 lakh

Rs.24 lakh

Q12. If $2^{n-1} = 256$, then 'n' is

44

12

8

9

Q13. A man and his wife plan for dinner but both are in different places. The probability of a man arriving first to the spot is 2/3 and his wife is 1/3. Find the probability that the wife reaches the spot first.

1/9

1/2

2/5

none of these

Q14. Three unbiased coins are tossed. What is the probability of getting at most two heads?

3/4

1/4

3/8

7/8

Q15. The sum of the first 16 terms of an AP whose first term and the third term are 5 and 15 respectively is

600

765

640

680

Q16. The arithmetic mean between two numbers is 75 and their geometric mean is 21. Find the numbers.

133 and 17

133 and 7

147 and 3

77 and 73

Q17. An alloy contains copper and aluminium in a ratio of 7:4. While making weapons from this alloy, 12% of the alloy gets destroyed. If there is 12 kg of aluminium in the weapon, then the weight of the alloy required is :

48 kg

40 kg

37.5 kg

14.4 kg

Q18. A vessel contains 60 litre of wine, 12 litres of wine taken out from it and replaced by water. Then again from mixture. 12 litres is again taken out replaced by water. The ratio of wine and water in the resultant mixture.

16 : 9

15 : 10

16 : 10

9 : 5

Q19. The average age of boys in a class of 30 is 15 years. If 10 more boys join the class, the average of the whole class is reduced by 1 year. What is the average age of newcomers?

12

11

15

20

Q20. The average salary of the entire staff in an office is Rs 250 per month. The average salary of officers is Rs 520 and that of non-officers is Rs. 200. If the number of officers is 15, then find the number of non–officers in the office.

83

81

87

56

Q21. In the question below, four statements are given, followed by conclusions: I, II and III. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Read the conclusions and then decide which of the given conclusions logically follows from the given statements disregarding commonly known facts.

Statements:
Some pens are keys
Some keys are locks
All locks are cards
No card is a paper

Conclusions:
I. No lock is paper.
II. Some cards are keys
III. Some keys are not paper.

Only conclusion I is true

Only conclusions II is true

Both conclusion I and II are true

All conclusions follow

Q22. **In the following question, three statements and three conclusions are given. Choose the option that shows the conclusions that logically follow from the given statements, disregarding commonly known facts.**

Statements :
All W's are M's
All Y's are M's
Only a few M's are Z's

Conclusions :
I. All Z's are Y's
II. Some Y's are W's is a possibility
III. Some M's are not Z's

All the conclusions follows

Only II follows

Only I and III follow

Only III follows

Only II and III follows

Q23. **Directions:**
What will come in place of question mark (?) in the following series?

32, 35, 64, 71, 104, ?, 152.

Q24. **Directions:**
What will come in place of question mark (?) in the following series?

14, 33, 71, 128, 204, ?.

Q25. **Directions:**
What will come in place of question mark (?) in the following series?

12, 6, 15, 67.5, ?, 3729.375

Q26. **Complete the series:**

CYD, FTH, IOL, LJP,?

- OFT
- LET
- OEK
- OET

Q27. Choose the correct order of the given alternative :
m n o n o p q o p q r s _ _ _ _ _

- mnopq
- oqrst
- pqrst
- qrstu

Q28. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.**

Statement: X=Y<W =Z ≥U;Z<P;Z<T

Conclusions:
I. W > P
II. T > U

- if only conclusion 1 is true
- if only conclusion 2 is true
- if either conclusion 1 or conclusion 2 is true

if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q29. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.**

Statement: G≥ K, K ≤ S, S = M, M < N

Conclusion:
I. N > K
II. G < S

if only conclusion 1 is true

if only conclusion 2 is true

if either conclusion 1 or conclusion 2 is true

if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q30. The question below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions Read both the statements and give an answer:

Jay put 12 cards on a table, some face up and the rest facedown. How many were put facedown?
(A) Jay put an even number of the cards facedown.
(B) Jay put twice as many of the cards face up as he put facedown.

if the question can be answered by using statement A alone but not by using B alone.

if the question can be answered by using statement B alone but not by using A alone.

if the question can be answered by using either statement alone

if the question can be answered by using both the statements together but not by either statement.

Q31. **The question below consists of a question and two statements numbered (I) and (II) given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and answer.**

In which year was Rahul born?
I) Rahul at present is 25 years younger than his mother.
II) Rahul's brother, who was born in 1964, is 35 years younger than his mother.

I alone is sufficient while II alone is not sufficient

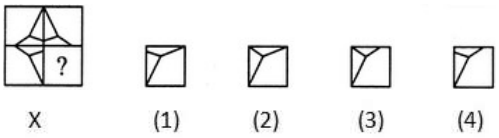
II alone is sufficient while I alone is not sufficient

Either I or II is sufficient

Neither I nor II is sufficient

Both I and II are sufficient

Q32. Identify the figure that completes the pattern.



1

2

3

4

Q33. **Fill in the blank with a suitable preposition:**

The police are responsible _____ the maintenance of law and order.

of

for

about

in

Q34. **Fill in the blank with a suitable article from the options given.**

When Jennie was young she knew _____ Picasso.

a

an

the

no article

Q35. **Correct the grammatical error which was in bold:**

Each of the boys in the class **were given the task** but none of them could finish the same on time.

I. Being given the task
II. Was given the task
III. Been given the task

- Both I and II
- Only II
- Only III
- Both I and III
- No correction required

Q36. **You are requested to identify the best way of writing the sentence in the context of the correct usage of standard written English. While doing so, you have to ensure that the message being conveyed remains the same in all the cases.**

The Greeks thought that the halcyon, or kingfisher, nested on the sea. All birds nest on land.

- Whereupon all birds nest on land, the Greeks thought that the halcyon, or kingfisher, nested on the sea.
- The Greeks thought that the halcyon, or kingfisher, nested on the sea, whereas all birds nest on land.
- Whenever all birds nest on land, the Greeks thought that the halcyon, or kingfisher, nested on the sea.
- The Greeks thought that the halcyon, or kingfisher, nested on the sea, as all birds nest on land.

Q37. Pick out the word most similar in meaning

Exempt

- duty
- provide
- relieve
- forgive

Q38. Select the option that is most nearly **OPPOSITE** in meaning to the given word

DILATORY (OPPOSITE)

narrowing

prompt

enlarging

portentous

sour

Q39. **Find out the antonym for the given word**

OSTENTATIOUS

showy

radical

modest

intrinsic

Q40. **In the given question, a sentence has been broken into five parts, and the parts are jumbled. Choose the option that provides the correct way to rearrange the parts to form a grammatically and contextually meaningful sentence.**

A. the Indigenous group belonging to the Mexican state of Sonora
B. this camp was established to safeguard
C. significance they hold for the Comcaac people,
D. olive ridley turtles, not only because they are
E. endangered, but also because of the cultural

BCEAD

BDECA

ACDBE

DBCAE

Q41. **In the following question, statements 1 and 6 are respectively the first and the last sentences of a paragraph, and statements A, B, C, and D come in between them. Rearrange A, B, C, and D in such a way that they make a coherent paragraph together with statements 1 and 6. Select the correct order from the given choices and mark its number as your answer.**

1. A lot of us do not realize that to carry out our day-to-day functions we need a minimum amount of endurance strength and flexibility in our bodies.
A. Adding to this, the 21st century seems to have many unforeseen emergencies in store for all of us; whether it be the 9/11 attack, an outbreak of epidemics, or strikes of terrorism - one never knows where and when what might happen.
B. Our current ways of life in general, are no longer providing our bodies with sufficient exercise to maintain adequate levels of fitness.
C. The realization of this dawns only when the necessity occurs, and that could be too late.

D. When we give in to the comforts of modern-day living and inadvertently let our bodies deteriorate, our abilities to carry out physical activities are so adversely affected that we may not derive the full pleasure of living.
6. In such circumstances proving the quality and most likely the longevity of our lives is a matter of personal choice, and depends greatly on how well we are able to pursue sound physical fitness programmes that help maintain a strong body and mind.

- DCBA
- BADC
- ABCD
- DACB

Q42. **From the given options choose the one that best expresses the given sentence in active/passive voice:**

The oldest house in town is being restored by the Historical Society.

- The oldest house in town was being restored by the Historical Society
- The oldest house in town had been restored by the Historical Society.
- The Historical Society is restoring the oldest house in town.
- The oldest house in town has been restored by the Historical Society.

Q43. **Change from direct speech to indirect speech**

"Hurry up," she said to us.

- She said us to hurry up.
- She says us to hurry up.
- She told us to hurry up.
- She tell us to hurry up.

Common Content:

Direction : Study the following information carefully and answer the following questions.

Ten persons are sitting in two parallel rows and facing each other. P, Q, R, S, T sit in row I facing south and J, K, L, M and N sit in row 2 facing north.
P sits third from one of the ends and faces the one who sits second to the left of L. One person sits between P and Q who doesn't face L. Two persons sit between Q and R. Two persons sit between J and M and one of them sits at one of the extreme ends. S who doesn't sit at any of the ends faces the one who sits immediate left of K. Immediate neighbors of S doesn't face M.

Q44. If J is related to P and K is related to T then N is related to ?

- S

R

P

T

Q45. Who among the following person faces K?

Q

R

T

P

Q46. Who among the following sits at extreme ends?

N

M

S

T

Common Content:

A + B means A is the wife of B.
A – B means A is the daughter of B.
A × B means A is the sister of B.
A / B means B is the son of A.
A @ B means A is the brother of B.

Q47. If F × B + Z @ C – R + P is given, then how is P related to B?

Uncle

Aunt

Mother-in-law

Father-in-law

Q48. If F × B + Z @ C – R + P is given, how is R related to C?

Son

Daughter

Mother

Cannot be determined

Q49. If G + B/C/D @ H is given, then how is D related to G?

Grandmother

Grandfather

Grandson

Son

Common Content:

Direction: Study the following information and answer the given questions.

Point B is 12 m South of point A. Point C is 24 m East of point B. Point D is 8 m South of point C. Point D is 12 m East of point E and point F is 8 m North of point E.

Q50. If a man is standing facing North at point C, how far and in which direction is he from point F?

12m east

24m west

12m west

24m east

Q51. What is the direction of D with respect to B?

south west

south east

south

none of these

Common Content:

Read the passage and answer the following questions:

Climate change occurs when there is a build-up of carbon in the atmosphere. However, if carbon is collected (pr sequestered) in the soil and used to increase productivity, the results are three-fold - a reduction in atmospheric carbon levels, increased crop yields, and more income for farmers. Ironic as it may sound, in dry land agriculture the availability of organic carbon in the soil is the key to increasing productivity. Hence, "look down at the soil to make your agricultural productivity look up" is the new motto of agricultural scientists, especially those working with natural resource management and watershed development. This, however, calls for massive investment by developed and developing countries to support dryland agricultural research and corresponding support systems. Although there is much attention and support being given to irrigated agriculture, drylands occupy 80 percent of agricultural land and generate 58 percent of the world's staple foods. In fact, most of the food for the poor in developing countries is produced in dryland and rain-fed areas.

A close look at the rain-fed regions shows a grim picture of water scarcity, fragile environments, drought, and land degradation caused by low rainwater use efficiency, high population pressure, poverty, poor infrastructure, and inappropriate policies. Drought and land degradation are interlinked in a cause-and-effect relationship, which in turn causes poverty. This unholy nexus among drought, poverty, and land degradation must be broken if we have to meet the Millennium Development Goal of halving the number of food-insecure poor by 2015. Added to this is the threat of climate change. It is anticipated that climate change will lead to more rainfall variability and increased frequency of extreme events, resulting in more and longer dry spells due to global warming. Innovative strategies are required to help the dry land farmer overcome these difficulties. The world's top research center for dryland agriculture, the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) is working on increasing the organic carbon level in the soil through increased sequestration (accumulation from the environment).

Soil organic carbon is the source of energy to fuel biological activities in the soil, which in turn controls the availability of nutrients for plant growth as well as soil water availability - a critical yield component in tropical dry lands. Increased carbon sequestration cannot only increase nutrient supply and turnover capacity of the soil but also result in substantial changes in the physical and biological properties of the soil, which increase water holding capacity and the ability to cope with longer dry spells during crop growth. ICRISAT's research has shown that legumes in general and pigeon peas, in particular, have the ability to sequester more carbon in tropical soils. Through the adoption of improved water, soil, and crop management options, carbon sequestration in legume-based systems can be improved further. If this is adopted within an integrated community watershed, the productivity of rainfed agriculture can be doubled, providing increased incomes to farmers.

Q52. One of the main reasons for climate change, according to the passage, is

- the increase in the production of carbon.
- the non-availability of organic carbon in the soil.
- the increased accumulation of carbon in the atmosphere.
- the increased accumulation of carbon in the soil.

Q53. Agricultural productivity in dry lands can be increased if

- massive investments are made by the developed countries.
- the availability of organic carbon in the soil is increased.
- the scientists start working on natural resource management.
- watershed development projects are given more financial support by the government.

Q54. The Millennium Development Goal stated in the passage aims at

- breaking the network among poverty, drought and land degradation by 2015.
- halving the number of food-insecure poor by 2015.
- calling for, massive investments by developed countries to support dry land agricultural research.
- increasing agricultural productivity two-fold to meet the growing population.

Q55. Which of the following is NOT a reason for the dreadful environmental conditions existing in all rainfed areas?

- Low rainwater use efficiency.
- High population pressure.
- Poor infrastructure.
- Lack of support for agricultural research.

Q56. Which of the following is NOT TRUE, according to the passage?

- Nearly 58 per cent of the world's staple food is produced from dry lands.
- An increased carbon level in soil can result in an increase in crop yields.
- Scientists working on natural resource management and watershed development pay more attention to the soil and its carbon levels.
- The developing and the developed countries give massive financial support to dry land agricultural research.

Common Content:

Choose the appropriate words to complete the passage

The world is witnessing food price turbulence again. A bad drought in Russia, rising demand in the US and developing countries, and Pakistan's blighted crop prospects after its floods are keeping prices of commodities such as cereals, sugar, oil and meat high. The Food and Agriculture Organisations monthly food price (__1_) is heading north. India is not (___2___) from this problem even at the best of times. For the week that ended on 11_ September, food prices as (__3___) by the Wholesale Price Index rose by 15.86%.Given the robust demand for foodstuffs, a time of price volatility calls for a careful look at the design issues surrounding food supply management. At times, even huge food stocks are not able to (___ 4___) rising food prices. The fault lies in how food is released to traders by government agencies such as the Food Corporation of India (FCI). This problem is apart from FCIs high carrying cost of food grains. But this is not the problem at (____5____)

Q57. Choose the appropriate word for the blank numbered (5)

- juncture
- hand
- best
- this

Q58. Choose the appropriate word for the blank numbered (1)

- rate
- scheme

index

state

Q59. Choose the appropriate word for the blank numbered (4)

slow

find

identify

arrest

Q60. Choose the appropriate word for the blank numbered (2)

immune

separate

overcome

devoid

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

```
1 Integer a,b,c,d
2 Set a=30,b=40,c=10
3 d=(a*b)/c
4 Print d
```

40

120

100

70

Q2. What will be the output of the following pseudocode?

```
1  Integer j, m
2  Set m=1
3  Integer a[4] = { 0,2,3,3}
4  if (a[1]<1)
5     if ( a[2]<2)
6        m=a[3]
7     Else
8        m=a[1]
9     End if
10    m=m^a[1]
11 End if
12 Print m
```

5

0

1

12

Q3. What will be the output of the following pseudocode?

```
1  Integer i
2  Set i = 3
3  do
4     print i + 3
5     i = i - 1
6  while(i not equals 0)
7  end do while
```

6 6 6

6 5 6

5 5 5

6 5 4

Q4. What will be the output of the following pseudocode for a=4, b=3, and c=8?

```
1  Integer funn(Integer a, Integer b,Integer c)
2     for (each c from 5 to 9)
3        a=(c^5)^a
4        if(b^c^a)<(c&a&b)
5           b=(1&11)&c
6        Else
7           Continue
8        End if
9        b=(c+2)+a
10    End for
11 return a+b
```

16

0

7

19

Q5. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=3,b=4,c=5
3  if ((b-c+a)<(a-b)
4     a=(a+9)&a
5     b=(c&c)+a
6     if ((a+6)>(c-a))
7        c=(c+c)+c
8     End if
9     c=b+a
10    c=8+a
11 End if
12 Print a+b+c
```

16

8

22

12

Q6. What will be the output of the following pseudocode f(x) x =777

```
1  Integer fun(Integer x)
2     if (x > 12)
3        fun (x / 9)
4        print x
5     else
6        for (each i from 1 to 3)
7           x = x + i
8           print x
9        end for
10    end if
11 end Function fun()
```

11 13 14 85 777

12 14 15 86 777

10 12 15 86 777

10 12 13 85 777

Q7. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=3,b=2,c=10
3  for (each c from 3 to 7)
4     b=3+b
5     a=(11+1)*a
```

```
5     a=(11+1)^c
6   End for
7   c=6+a
8   a=b+c
9   for (each c from 3 to 4)
10    a=(b+12)+b
11  End for
12  Print a+b
```

- 60
- 68
- 72
- 63

Q8. What will be the output of the following pseudocode?

```
1   Integer a,b,c,d,i
2   Set a=1, b=0,c=6,d=7
3   for (each i from 0 to 3)
4     if (a||b||(c && d))
5       print a++
6     end if
7     if (a equals 2)
8       print a++
9     end if
10    if (a equals 1)
11      jump out of the loop
12    end if
13  end for
14  print "journey"
```

- 1 2 journey
- 1 2 3 4 5
- 1 2
- 1 2 3 4 5 journey

Q9. What will be the output of the following pseudocode?

```
1   Integer a,b,c
2   Set a=9,b=2,c=13
3   a=(a+2)+a
4   c=10&b
5   a=(c&c)+c
6   Print a+b+c
```

- 5
- 8
- 12

13

Q10. What will be the output of the following pseudocode?

```
Integer j, m
Set m=0
Integer a[4] = { 3,0,2,1}
for (each j from 0 to 3)
   if (a[j])
      m=a[j]
   End if
   if (j)
      m=a[j]+1
   End if
End for
Print m
```

2

5

1

8

Q11. What will be the output of the following pseudocode?

```
Integer pp, qq,rr
Set pp=6, qq=8, rr=8
for(each rr from 2 to 3)
 if((pp-rr)<(rr+pp))
   qq=qq+pp
 End if
 qq=qq+pp
 pp=10&pp
End for
Print pp+qq
```

22

37

30

26

Q12. What will be the output of the following pseudocode?

```
Integer a, b, c
Set a=0, b=9, c=5
a=(c+12)^b
if((b+c)>(c-b) || 3>b)
  b= (11+5)&b
End if
b=1&c
Print a+b+c
```

40

30

31

25

Q13. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=9, q=6, r=5
3  p=(r&5)^r
4  p=(q+q) + q
5  q = 10 ^p
6  Print p+q+r
```

47

46

57

56

Q14. What will be the output of the following pseudocode?

```
1  Integer arr[] = {10,7,2,6,1,2,3}
2  Integer sum , x
3  Set sum =0
4  for(each x from 0 to 6)
5    if(arr[-1] mod 3 >1)
6      sum = sum +arr[x]
7    End if
8  End for
9  Print arr[4]
```

4

9

1

5

Q15. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=9, q=10, r=6
3  q=8+q
4  if((p>r || ((q+r) > (r-q)))
5     p=(p&12)+p
6  End if
7  p=(r+4)^r
8  Print p+q+r
```

37

36

44

32

Q16. What will be the output of the following pseudocode if the value of n=5 and v=1, 2, 3, 4, 5?

```
1 Integer n, s, c, v
2 Set n=5, s=0
3 for(each c from 1 to n)
4     Read the value of v
5     s=s+v
6     s=s+n
7 end for
8 Print s
```

22

25

45

40

Q17. What will be the output of the following pseudocode?

```
1 Integer a , b, c
2 Set a=1, b=4, c=8
3 if(b>c OR (a+b)<(b-a))
4    a= (7+10) + a
5    b= (8+5) + a
6 End if
7 Print a + b + c
```

13

26

17

8

Q18. What will be the output of the following pseudocode?

```
1 Integer p,q,r
2 Set p=3, q=8, r=10
3 for(each r from 3 to 7)
4    q= (11+9)+q
5    p= 12+p
6 End for
```

```
7  for(each r from 2 to 4)
8    q=(7+4)^q
9  End for
10 Print p+q
```

- 183
- 172
- 160
- 166

Q19. What will be the output of the following pseudocode?

```
1  integer a1, p1
2  set a1= 15
3  a1= a1- 1
4  a1= a1 * 2 + 8
5  a1= a1 / 2 + 9 * 5
6  p1 = a1 / 5 + 6
7  print p1
```

- 18
- 23
- 19
- 3

Q20. DNS uses which transport protocol for communication?

- TCP
- UDP
- ICMP
- ARP

Q21. What is the primary function of SMTP?

- File Transfer
- Email Transmission
- Name Resolution

Remote Login

Q22. The default port for HTTP is ____

21

25

80

110

Q23. FTP (File Transfer Protocol) is based on which model?

Connectionless model

Connection-oriented model

Client-server model

Peer-to-peer model

Q24. Which protocol is used for email transmission?

HTTP

SMTP

FTP

DNS

Q25. Which topology is commonly used in token ring networks?

Bus

Star

Ring

Mesh

Q26. Which of the following is an example of Bluetooth?

- Wide area network
- Virtual private network
- Local area network
- Personal area network

Q27. Which topology is most commonly used in home networks?

- Ring
- Star
- Mesh
- Bus

Q28. Which topology provides fault tolerance and redundancy by having multiple paths between nodes?

- Bus
- Ring
- Mesh
- Star

Q29. Which organization is responsible for the development and maintenance of the SSL/TLS protocols?

- W3C
- IETF
- ISO
- ITU-T

Q30. What is the purpose of the Multipurpose Internet Mail Extensions (MIME) standard?

- To encode binary data for transmission in email messages
- To encrypt email messages for security

To authenticate email senders and receivers

To limit the size of email attachments

Answer Key & Solution

Section 1 - Aptitude

Q1

7: 40 am

Solution

Note: Assume the total work = LCM of the given time

Take the LCM of hours = LCM of (8, 12, and 24) = 24

Let the total work = 24

Note: One hour work = (total work/ time)

Now,

A's one hour work = 24/8 = 3 unit

B's one hour work = 24/12 = 2 unit

C's one hour work = 24/24= 1 unit

ATQ, between 3 am - 4 am, A works alone and finish 3 units of work

Between 4 am - 5 am, A and B work and complete 3+2 = 5 units of work

Between 5 am - 6 am, A, B, and C works and finishes 3+2+1 = 6 units of work

Or, between 3 am to 5 am, 8 unit of work is finished.

Remaining 24-8= 16 unit (till 5am)

After 5 am, A+B+C work together and finish 6 unit of work per hours.

i.e., 6 unit work = 1 hour

To complete the 16 unit work, multiply both sides with 2

i.e., 12 unit work = 2 hours

Remaining work = 16-12 = 4 unit work

As we know, 6 work = 1 hour

Or, 1 work = 1/6 hours

Or, 4 work = 4/6 or 2/3 hours.

That means (2 + 2/3) hours requires after 5 am.

So, time taken to complete the work is (5 +2 + 2/3) = 7:40am

Q2

P = Rs. 440; Q = Rs. 385; R = Rs. 280

Solution

P= 7 days

Q= 8 days

R= 11 days

LCM= Total work= 7*8*11= 616 units

Efficiency of P= 88 units/day

Efficiency of Q= 77 units/day

Efficiency of R= 56 units/day

P share= (88/221)*Rs.1105= 440

Q share=(77/221)*Rs.1105= 385

R share = (56/221)*Rs.1105= 280

Q3

30

Solution

Speed= (200/24)*(18/5)= 30 km/hr

Q4

10 kmph

Solution

Speed of the steamer downstream Sd = 30/2 = 15 kmph

Speed of the steamer upstream Su = 30/6 = 5 kmph

Let speed of the steamer be ‘x’, then speed of the current would be (x/2).

According to the question,

x + (x/2) = 15 ……..(1)

x − (x/2) = 5 ……..(2)

From equations (1) and (2) we get,

2x = 20 ⇒ x = 10 kmph

Hence, speed of the steamer is 10 kmph.

Speed of steamer in still water = (speed downstream + speed upstream)/2

= (15 + 5)/2 = 10 kmph.

Hence, speed of the steamer in still water is 10 kmph.

Q5

53%

Solution

CP = Rs 100, P = 30.05%

S.P. = Rs 130.05

M.P. = 130.05 × 100/90 x 100/85 x 100/85 = Rs 200

SP= 200 x 90/100 x 85/100 = Rs. 153

So profit=53%

Q6

56.8%

Solution

Successive discount of 10% and 20% = -10 -20 + (-20*-10)/100 = -28%

Successive discounts of 10%, 20% and 40% = -28 -40 +(-28*-40)/100 = -56.8%

Q7

92 $\frac{1}{2}$

Solution

90 kg = 81 kg sugar 9 kg maida

Adding = 30 kg sugar

Total = 111kg sugar 9kg maida

Hence, (new) mixture will have = 111/120 × 100 = 92.5 = 92 1 /2

Q8

25

Solution

Original price = Rs. 30

increased price = Rs. 37.5

increase in price = 37.5 - 30 = 7.5

increased percentage = (increase price / original price) ×100

= (7.5/30)×100 = 25

Q9

9 years

Solution

$A= P(1 + R/100)^N$

Given amount double in 3 years.

$2P= P(1 + R/100)^3$

$(1 + R/100)^3 = 2$

Let Amount becomes 8 times in N years,

$8P= P(1+ R/100)^N$

$(1+ R/100)^N= 8 = 2^3= ((1 + R/100)^3)^3= (1+ R/100)^9$

N= 9 years

Q10

14641

Solution

$A = 10000(1+20/200)^4$

= 14641

Q11

Rs.24 lakh

Solution

Consider Alphonsa,s total property = x lakh

Share of his wife = $\frac{x}{2}$ lakh

Share of his each son = $\frac{x}{8}$ lakh

After the death of Ben , Carl &Dave get = $\frac{x}{24}$ lakh

Now property Carl & Dave = $\frac{5x}{24}$ lakh

After the death of Carl, dave's property becomes $\frac{15x}{48}$ lakh.

After the death of dave, his mother property becomes $\frac{15x}{96} + \frac{x}{2} = \frac{63x}{96}$ lakh

$\frac{63x}{96}$ = 15,75,000

x = 24 lakh.

Q12

9

Solution

We have, $2^{n-1} = 256 = 2^8$. Hence n – 1 = 8. Therefore n = 9.

Q13

1/9

Solution

Man's probability is 2/3

Wife's probability is 1/3

Man not reaching probability = 1- 2/3 =1/3

If the wife has to reach first then probability = wife's probability of reaching first x man not reaching probability

= 1/3 x 1/3 =1/9

Q14

7/8

Solution

Here S = [TTT, TTH, THT, HTT, THH, HTH, HHT, HHH]

Let E = event of getting at most two heads

Then, E = {TTT, TTH, THT, HTT, THH, HTH, HHT}

∴P(E)=n(E) / n(S)

=7 / 8

Q15

680

Solution

a=5

a+(n-1)d = 15

a+2d=15

d=5

t16=80

S16= (a + l)(n/2) = 680

Q16

147 and 3

Solution

Let the required numbers are ‘a’ and ‘b’.

We have = 75 ⇒ $\frac{a + b}{2}$ =a+b = 150 ....(i)

Also sq(ab) 21 or ab = 441

We know that $(a-b)^2 = (a+b)^2 - 4ab = 150^2 - 4 \times 441 = 22500 - 1764 = 20736$

⇒ a – b = 144 ….(ii)

Adding (i) and (ii), we get 2a = 294 ⇒ a = 147

(i) ⇒ b = 3.

So the numbers are 147 and 3.

Q17

37.5 kg

Solution

Since 12% of alloy itself is destroyed, the ratio will not change.

Copper : Aluminium = 7 : 4

Copper = 21 kg and Aluminium = 12 kg Total = 33 kg

88%= 33 Kg

100% = 33/0.88 = 37.5 kg

Q18

16 : 9

Solution

Final quantity of Wine= $60(1- (12/60))^2$

= $60(4/5)^2$

= 60*4*4/5*5

=38.4 lit

Quantity of Water= 60 - 38.4= 21.6 lit

Ratio of Wine: Water= 38.4:21.6= 16:9

Q19

11

Solution

Average of 30 boys= 15

Total of 30 boys= 15*30= 450

Let average of 10 new comers= x

Now average is 14

450+10x/(40)= 14

Solving we get x= 11

Q20

81

Solution

Let the required number of non–officers = x

200x + 520 x 15 = 250 (15 + x)

250x – 200x = 520 * 15 – 250 x 15

50x = 4050

x = 81

Q21

All conclusions follow

Solution



All follows

Q22

Only II and III follows

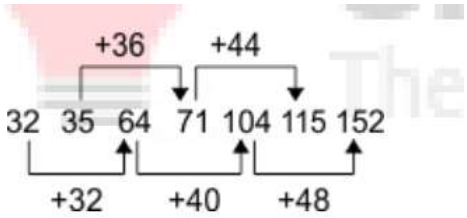
Solution

No Solution

Q23

115

Solution



Q24

None

Solution

Series Pattern Given Series

14 14

14 + 19 = 33 33

33 + 19 × 2 = 71 71

71 + 19 × 3 = 128 128

128 + 19 × 4 = 204 204

204 + 19 × 5 = 299 299

Q25

438.75

Solution

Series Pattern Given Series

12 12

12 × 0.5 = 6 6

6 × 2.5 = 15 15

15 × 4.5 = 67.5 67.5

67.5 × 6.5 = 573.75 438.75

438.75 × 8.5 = 6024.375 3729.375

Q26 OET

Solution

CYD, FTH, IOL, LJP

First letter: C + 3 = F, F + 3 = I, I + 3 = L

Therefore, L + 3 = O

Second Letter: Y - 5 = T, T - 5 = O, O - 5 = J

Therefore, J - 5 = E

Third Letter: D + 4 = H, H + 4 = L, L + 4 = P

Therefore, P + 4 = T

The next set is OET.

Q27 pqrst

Solution

The series is mno/nopq/opqrs/pqrst.

Q28 if only conclusion 2 is true

Solution

Combining the inequalities; X=Y<W =Z ≥U X=Y<W =Z<P X=Y<W =Z<T

W < P and T>Z ≥U, thus, T > U. II is true.

Q29 if only conclusion 1 is true

Solution

Combined Inequalities: G ≥ K ≤ S = M < N G≥ K ≤ S = M < N, K< N.

Hence, conclusion I is true. G≥ K ≤ S = M < N, No relationship cannot be established between G and S. Hence, conclusion II is not true.

Q30

if the question can be answered by using statement B alone but not by using A alone.

Solution

Statement(A): If X denotes the number of cards that were put facedown, it follows from (A) that X could be 2,4,6,8,10, or 12. Thus statement (A) alone is not sufficient.

Statement (B) it can be determined that 2X + X = 3X = 12, or X = 4. Therefore, statement (B) alone is sufficient.

Q31

Both I and II are sufficient

Solution

From both I and II, we find that Rahul is (35 - 25) = 10 years older than his brother, who was born in 1964. So, Rahul was born in 1954.

Q32

4

Solution



Q33

for

Solution

responsible is usually followed by the preposition for

Q34

no article

Solution

Proper nouns will not take an article

Q35

Only II

Solution

There is the error in the subject-verb agreement here because the subject here is in singular form whereas the corresponding verb has been used in the plural form i.e. were given. The subject *each* is singular and is followed by singular verbs only. Therefore it should have been *was given.*

The correct sentence would have been:

Each of the boys in the class **was given the task** but none of them could finish the same on time.

This makes Option B the correct choice among the given options.

Q36

The Greeks thought that the halcyon, or kingfisher, nested on the sea, whereas all birds nest on land.

Solution

Whereas (in choice b) is the logical subordinator, establishing contrast.

Q37

relieve

Solution

Exempt means free from, or relieve.

Q38

prompt

Solution

Dilatory means unhurried or slow. Portentous means warning. Sour means bitter. Prompt means fast.Prompt is the antonym.

Q39

modest

Solution

Ostentatious means characterized by or given to pretentious or conspicuous show in an attempt to impress others

Modest means free from ostentation or showy extravagance

Q40

BDECA

Solution

(B) mentions that this camp was established to safeguard something.

(D) starts with olive ridley turtles who were to be safeguarded. (E) mentions that they are endangered.

D and E can be linked by the conjunction pair 'not only - but also'. So, the pair D-E states that they are to be safeguarded not only because they are endangered but also because they hold 'cultural' significance.

(C) follows (E) because (E) ends with an adjective 'cultural' which can be followed by a noun only, i.e., 'significance'.

Hence, BDECA is the correct arrangement

Q41

DCBA

Solution

Statement 1 says to carry out our day-to-day activities a minimum amount of endurance and flexibility in our bodies is needed. If we look at the statements, 'D' is the most appropriate one to follow 1. 'D' emphasizes on the fact that our physical activities are adversely affected, which is a continuation of the idea expressed in '1'. The words "the realization of this ...." refers to the reason for our abilities to be adversely affected. Hence 'C' is a better statement to follow 'D'. Further in 'B' we find emphasis on the current ways of life. Thus 'DCB' is the logical order which is further followed by 'A'. Choice (1)

Q42 The Historical Society is restoring the oldest house in town.

Solution

Solution:

The given sentence is in passive voice.

When changed to active voice, is being restored becomes is restoring

The right answer is "The Historical Society is restoring the oldest house in town ."

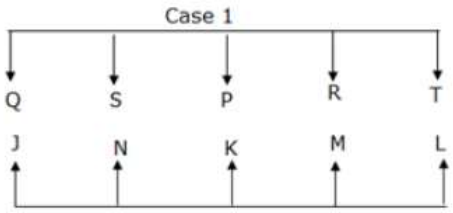
Q43 She told us to hurry up.

Solution

You say words. You tell someone about something. So the right answer is "She told us to hurry up".

Q44. R

Solution



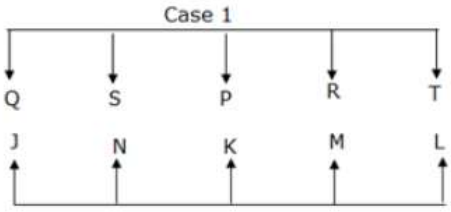
P is opposite to the person sitting second to right of J

T is opposite to the person sitting second to right of K

So R is opposite to person sitting second to right of N

Q45. P

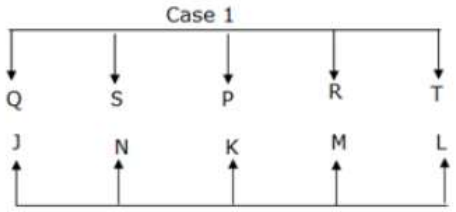
Solution



rom the diagram P faces K

Q46. T

Solution



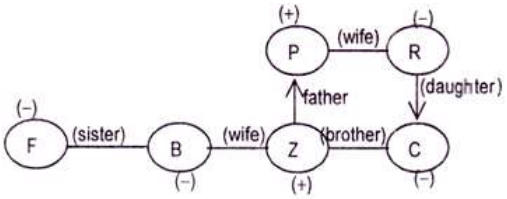
T sits at the extreme end.

Q47. Father-in-law

Solution

Given: F*B + Z@C − R + P

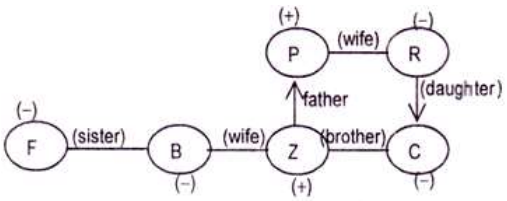
It can be represented as shown



From the above diagram it is clear that, P is the father-in-law of B.

Q48. Mother

Solution

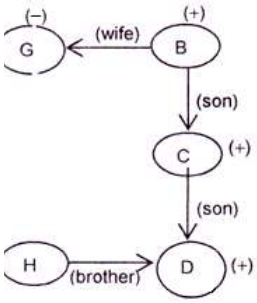


Q49. Grandson

lution

iven : G + B / C / D @ H

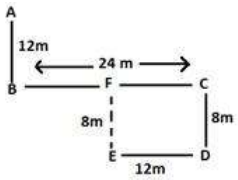
/e can have the following diagram



rom the figure it is clear that, D is the grand son of G.

Q50. 12m east

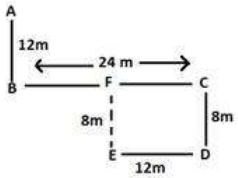
Solution



He is 12 m east from F

Q51. south east

Solution



D is in south east to B

Q52. the increased accumulation of carbon in the atmosphere.

Solution

Refer to the first sentence of para 1 which talks about the 'build-up of carbon in the atmosphere. The right answer is "the increased accumulation of carbon in the atmosphere."

Q53. the availability of organic carbon in the soil is increased.

Solution

Refer to the third-sentence of the first paragraph, according to which the 'key to increasing productivity' is 'the availability of organic carbon in the soil'. The right answer is "the availability of organic carbon in the soil is increased."

Q54. halving the number of food-insecure poor by 2015.

Solution

Refer to the third sentence of para two which mentions the goal. The right answer is "halving the number of food-insecure poor by 2015."

Q55. Lack of support for agricultural research.

Solution

Refer to the first sentence of para two which does not mention 'lack of support for agricultural research'.The right answer is "Lack of support for agricultural research."

Q56. The developing and the developed countries give massive financial support to dry land agricultural research.

Solution

Choice "Nearly 58 per cent of the world's staple food is produced from dry lands." is true according to the penultimate sentence of para one. Choices "An increased carbon level in soil can result in an increase in crop yields." &
"Scientists working on natural resource management and watershed development pay more attention to the soil and its carbon levels." are stated In the second sentence and the fifth sentence of para one. The right answer is "The developing and the developed countries give massive financial support to dry land agricultural research."

Q57. hand

Solution

The idiomatic phrase , not the problem at hand completes the sentence.

Q58. index

Solution

idex is the correct option, since it represents variety of items as one entity.

Q59. arrest

Solution

According to the passage the food stocks are not able to stop or control , the word from the option that coveys this meaning is arrest.

Q60. immune

Solution

immune refers to being resistant

Section 2 - Pseudocode and Computer Fundamentals

Q1

120

Solution

d=(30*40)/10 =>120

Q2

1

Solution

If condition gets failed so 1 gets printed.

Q3

6 5 4

Solution

Initially, i=3.

Iteration 1:

prints 6 and update i=2

Iteration 2:

prints 5 and update i=1

Iteration 3:

prints 4 and update i=0

When I equals 0, the loop is terminated.

Q4

7

Solution

Else block gets execute in all the iteration and 'a' updated as 4,7,5,8,4. So finally 4+3 =7 gets printed.

Q5

12

Solution

Initially a=3,b=4,c=5, If block condition gets failed because 2 < -1 (False). So 3+4+5 = 12.

Q6

10 12 15 86 777

Solution

No Solution

Q7

63

Solution

When c=3 , b=3+2=>5 and a=(11+1)*3=36 same process repeated till c= 7 (finally a=84 and b=17)

In Second for loop (line 9) for all iteration a=17+12+17=>63

Q8

1 2 3 4 5 journey

Solution

When i=0, 1st and 2nd if block gets executed by printing 'a' i.e. 1,2 .('a' incremented in each block)

When i=1,2,3 for all iteration 1st if block gets executed by printing 'a' i.e. 3,4,5 .('a' incremented in each block) and finally "journey" gets printed.

Q9

8

Solution

a=9+2+9 =20

c=10&2=2

a=(2&2)+2=4

So 4+2+2=8

Q10

2

Solution

First if block executes when j=0,2,3 and second if block executes when j=1,2,3. In second if block finally m=a[3]+1 i.e. 2 gets printed.

Q11

26

Solution

In each iteration if block gets executed and update value of pp and qq. Finally pp=2 and qq=24, so 26 gets printed.

Q12

30

Solution

If block gets executed and updated the value of b. Finally a=24,b=1,c=5. so output is 30.

Q13

47

Solution

Initially p=9, q=6, r=5,

p=(5&5)^5=>0

p=(6+6) + 6 =>18

q = 10 ^18=> 24. So 18+24+5=>47

Q14

1

Solution

Array values are arr= [10,7,2,6,1,2,3] where arr[4]=1 gets printed.

Q15

36

Solution

No Solution

Q16

40

Solution

Inside for loop sum of 5 natural number is done and for each iteration 5 added up to variable s i.e. 1+2+3+4+5= 15 and five added 5 times so 25. So the output is 25+15=40.

Q17

13

Solution

Initially a=1

b=4

c=8

If condition gets failed so 1+4+8= 13.

Q18

166

Solution

Inside first for loop:

When r=3,p and q updated as 15,28

When r=4,p and q updated as 27,48

When r=5,p and q updated as 39, 68

When r=6,p and q updated as 51, 88

When r=7,p and q updated as 63,108 and loop terminates.

Inside second for loop:

When r=2,q updated to 103

When r=3,q updated to 108

When r=4,q updated to 103 and loop terminates.

Finally, 63+103 = 166 gets printed.

Q19

18

Solution

Hint 1The value of a1 is initialized as 15Then a1 is subtracted by 1, it becomes 14Next a1 is multiplied by 2 and added with 8 so it becomes 36Next a1 is divided by 2 and added with the result of 9 * 5, it becomes 63then 63 divided by 5 it becomes 12 then added with 6, it becomes 18 which will be assigned to p1. Finally, when we print p1, it prints 18 as output

Q20

UDP

Solution

No Solution

Q21

Email Transmission

Solution

No Solution

Q22

80

Solution

No Solution

Q23 Client-server model

Solution

No Solution

Q24 SMTP

Solution

No Solution

Q25 Ring

Solution

No Solution

Q26 Personal area network

Solution

Explanation: Bluetooth is a wireless technology used to create a wireless personal area network for data transfer up to a distance of 10 meters. It operates on 2.45 GHz frequency band for transmission.

Q27 Star

Solution

No Solution

Q28 Mesh

Solution

No Solution

Q29 IETF

Solution

No Solution

Q30 To encode binary data for transmission in email messages

Solution

No Solution